

2. Technical Implementation

This section describes how the CVD platform is implemented: the frontend SPA, the backend REST API, the database schema, the research/analytics workflow, and the deployment/runtime setup.

Contents

- * [Tech Stack](#)
- * [Criteria Documentation](#) - ADRs for evaluation criteria
- * [Deployment](#)

Solution Architecture

High-Level Architecture

[Diagram](#) - High-level architecture diagram

Additional diagrams: - [ER Diagram](#) - [Deployment Diagram](#)

System Components

Component	Description	Technology
Frontend	Single-page application (disease libr...	React + TypeScript + Vite
Backend	REST API for diseases/symptoms/risk f...	Node.js + TypeScript + Express
Database	Persistent, normalized content store ...	PostgreSQL 16 + Prisma
Cache	Optional read caching to reduce DB lo...	Redis 7 (ioredis client)
External Services	Error monitoring and performance trac...	Sentry

Data Flow

```
[User Action] ? [Frontend (React)] ? [HTTP Request] ? [Backend (Express)]
|
+--> [Cache (Redis)] --(hit)--> [R...
|
+--> (miss) [Prisma] ? [PostgreSQL...
```

[UI Update] ? [Frontend renders data] ?-----+

Locale-aware content is handled by selecting translations (EN/RU) at the API/service layer based on the locale query parameter.

Key Technical Decisions

Decision	Rationale	Alternatives Considered
React + Vite SPA	Fast iteration and simple deployment ...	Next.js SSR/ISR
Express + Prisma API	Lightweight REST API with type-safe D...	Fastify, NestJS, Python (FastAPI/Django)
PostgreSQL normalized schema + transl...	Data integrity, M:N modeling (with me...	Single-table MVP, MongoDB
Redis caching for read-heavy endpoints	Improves latency and reduces DB load ...	No cache, in-memory cache
Observability via Sentry + structured...	Practical debugging and error reporti...	Console-only logs, full APM suites

Criteria Documentation

The evaluation criteria ADRs are located in [criteria/](#) and include:

- * [Frontend](#)
- * [Backend](#)
- * [Database](#)
- * [Research](#)
- * [Deployment](#)
- * [Observability](#)
- * [Testing](#)